

LECTURE 20

Advanced Normalization: Functional Dependencies and BCNF

A formal proof system for FDs, and the normal form that catches what 3NF misses

CSC270 · Database Systems

COMSATS University Islamabad, Lahore Campus

From Axioms to the Strictest Normal Form

- 1 Armstrong's Axioms — reflexivity, augmentation, transitivity + derived rules
- 2 The closure of a functional dependency set, F^+
- 3 Attribute closure, X^+ — the algorithm, and how it finds candidate keys
- 4 Boyce-Codd Normal Form (BCNF) and how it differs from 3NF
- 5 Testing for BCNF using attribute closure
- 6 The classic 3NF-but-not-BCNF relation, and BCNF decomposition

Armstrong's Axioms

Sound (only ever derive FDs that genuinely hold) and **complete** (repeated application derives EVERY implied FD).
Working example — StudentAdvisor(StudentNo, StudentName, AdvisorNo, AdvisorName, DeptNo, DeptName):

Axiom	Rule	Example
Reflexivity	If $Y \subseteq X$, then $X \rightarrow Y$ (trivial)	$\{StudentNo, StudentName\} \rightarrow StudentNo$
Augmentation	If $X \rightarrow Y$, then $XZ \rightarrow YZ$	$StudentNo \rightarrow StudentName$ gives $\{StudentNo, DeptNo\} \rightarrow \{StudentName, DeptNo\}$
Transitivity	If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$	$StudentNo \rightarrow AdvisorNo, AdvisorNo \rightarrow AdvisorName \Rightarrow StudentNo \rightarrow AdvisorName$

These three are the formal foundation everything else in this lecture builds on.

Derived Rule (from the 3 axioms)	Statement
Union (additivity)	If $X \rightarrow Y$ and $X \rightarrow Z$, then $X \rightarrow YZ$
Decomposition (projectivity)	If $X \rightarrow YZ$, then $X \rightarrow Y$ and $X \rightarrow Z$ separately
Pseudotransitivity	If $X \rightarrow Y$ and $WY \rightarrow Z$, then $WX \rightarrow Z$

Deriving an FD with Armstrong's Axioms

GOAL

From $F = \{\text{StudentNo} \rightarrow \text{AdvisorNo}, \text{AdvisorNo} \rightarrow \text{AdvisorName}\}$, prove $\text{StudentNo} \rightarrow \text{AdvisorName}$.

Step	Statement	Justification
1	$\text{StudentNo} \rightarrow \text{AdvisorNo}$	Given (in F)
2	$\text{AdvisorNo} \rightarrow \text{AdvisorName}$	Given (in F)
3	$\text{StudentNo} \rightarrow \text{AdvisorName}$	Transitivity, steps 1 and 2
4	$\text{StudentNo}, \text{DeptNo} \rightarrow \text{AdvisorName}, \text{DeptNo}$	Augmentation of step 3 with DeptNo

WHY THIS MATTERS

Each line follows from earlier lines by one axiom. Attribute closure, on the next slides, reaches the same conclusions faster.

Closure of FDs (F^+) and Attribute Closure (X^+)

F^+ , the closure of F , is every FD derivable from F by repeatedly applying Armstrong's Axioms — usually far too large to enumerate directly.

NOTE

Attribute closure X^+ (under F) is the set of ALL attributes functionally determined by X — every attribute A such that $X \rightarrow A$ is in F^+ . This answers the one question actually needed over and over: “given X , what does it determine?”

Algorithm:

```
Result := X
repeat
  for each FD (LHS  $\rightarrow$  RHS) in  $F$ :
    if LHS  $\subseteq$  Result: Result := Result  $\cup$  RHS
until Result stops changing  $\rightarrow$  return Result
```

Finding keys: K^+ = all attributes means K is a superkey; minimal such K is a candidate key.

Computing $\{\text{StudentNo}\}^+$

Start: Result = $\{\text{StudentNo}\}$



Apply $\text{StudentNo} \rightarrow \text{StudentName}$, $\text{StudentNo} \rightarrow \text{AdvisorNo}$
Result = $\{\text{StudentNo}, \text{StudentName}, \text{AdvisorNo}\}$



Apply $\text{AdvisorNo} \rightarrow \text{AdvisorName}$, $\text{AdvisorNo} \rightarrow \text{DeptNo}$
AdvisorNo now $\subseteq$ Result (just added)



Apply $\text{DeptNo} \rightarrow \text{DeptName}$ — done, no FD left to apply
 $\{\text{StudentNo}\}^+ = \text{ALL 6 attributes of StudentAdvisor}$

TIP

$\{\text{StudentNo}\}^+$ is all of StudentAdvisor, so $\{\text{StudentNo}\}$ is a candidate key. $\{\text{AdvisorNo}\}^+$ misses StudentNo: not a superkey.

Finding Candidate Keys with Attribute Closure

METHOD

K is a superkey iff $K^+ = \text{all attributes}$. K is a candidate key if it is a superkey and no proper subset is.

Attribute set X	Closure X^+ (steps)	Result	Verdict
{StudentNo}	add StudentName, AdvisorNo, then AdvisorName, DeptNo, then DeptName	all 6 attributes	Candidate key
{AdvisorNo}	add AdvisorName, DeptNo, then DeptName	misses StudentNo, StudentName	Not a superkey
{StudentName}	no FD has StudentName on the left	StudentName only	Not a superkey

CHECK

Checking every single attribute: only {StudentNo} reaches all six, so it is the only candidate key.

Boyce-Codd Normal Form (BCNF)

NOTE

A relation R is in BCNF if, for EVERY non-trivial FD $X \rightarrow Y$ that holds in R, X is a superkey of R — with no exception.

	3NF	BCNF
Exception clause?	Yes, if X isn't a superkey but every attribute of Y is prime	None — X must always be a superkey
Relationship	Every BCNF relation is automatically 3NF	The reverse is NOT guaranteed

Compare directly against 3NF's definition (Lecture 19): BCNF removes 3NF's prime-attribute exception entirely — exactly the gap the next section makes concrete.

for each FD $(X \rightarrow Y)$ in F (or a minimal cover):
compute X^+ ; if $X^+ \neq$ all attributes of R $\rightarrow$ X not a superkey $\rightarrow$ VIOLATES BCNF

Applying the BCNF Test to the 3NF Schema

TEST

For each FD $X \rightarrow Y$: compute X^+ . If X^+ is not all of R, X is not a superkey, and the FD violates BCNF.

Relation	FD (determinant X)	X^+	X superkey?	BCNF?
Customer	$\text{custNo} \rightarrow \text{custName, custAddress}$	all 3 attributes	yes	OK
Order	$\text{orderNo} \rightarrow \text{orderDate, custNo}$	all 3 attributes	yes	OK
Product	$\text{productNo} \rightarrow \text{productName, unitPrice}$	all 3 attributes	yes	OK
OrderLine	$\{\text{orderNo, productNo}\} \rightarrow \text{qty}$	all 3 attributes	yes	OK

RESULT

Every determinant is a superkey, so the Lecture 19 schema is already in BCNF. No decomposition is needed.

Enrollment(StudentNo, CourseNo, InstructorNo)

Business rule: each instructor teaches only one course (in a given term).

StudentNo	CourseNo	InstructorNo
S1	CS201	I1
S2	CS201	I1
S3	CS305	I2

$$F = \{ \{ \text{StudentNo}, \text{CourseNo} \} \rightarrow \text{InstructorNo}, \text{InstructorNo} \rightarrow \text{CourseNo} \}$$

- {StudentNo,CourseNo}+: apply FD1 $\rightarrow$ adds InstructorNo $\rightarrow$ all 3 attrs. Superkey, minimal $\rightarrow$ Candidate key #1
- {StudentNo,InstructorNo}+: apply FD2 (InstructorNo $\rightarrow$ CourseNo) $\rightarrow$ adds CourseNo $\rightarrow$ all 3 attrs. Superkey, minimal $\rightarrow$ Candidate key #2

Two candidate keys — between them, StudentNo, CourseNo, and InstructorNo are ALL prime attributes.

Candidate Keys of Enrollment, Step by Step

METHOD

Compute X^+ for each candidate set. FDs: {StudentNo, CourseNo} $\rightarrow$ InstructorNo and InstructorNo $\rightarrow$ CourseNo.

Attribute set X	Closure steps	Result	Verdict
{StudentNo, CourseNo}	apply FD1: add InstructorNo	all 3 attributes	Candidate key #1
{StudentNo, InstructorNo}	apply FD2: add CourseNo	all 3 attributes	Candidate key #2
{CourseNo}	no FD has CourseNo alone on the left	CourseNo only	Not a superkey

CHECK

Both candidate keys are minimal, so StudentNo, CourseNo, and InstructorNo are all prime attributes.

The Anomalies That Survive 3NF

SETUP

Enrollment(StudentNo, CourseNo, InstructorNo) is in 3NF but not BCNF. The redundancy shows up in the rows:

Anomaly	Rows involved	Consequence
Modification	I1 teaches CS201 in rows for S1 and S2	Move I1 to CS410 in one row only: I1 appears to teach two courses
Insertion	I3 teaches CS500 but has no student yet	I3's course cannot be recorded until someone enrolls
Deletion	S3 is the only row for I2 and CS305	Deleting S3's enrollment erases that I2 teaches CS305

KEY IDEA

Each anomaly comes from InstructorNo → CourseNo, whose determinant is not a superkey. BCNF forbids that.

3NF Passes, BCNF Fails

Is Enrollment in 3NF?

Rule checked	Result
$\{\text{StudentNo}, \text{CourseNo}\} \rightarrow \text{InstructorNo}$	Determinant IS a candidate key ✓
$\text{InstructorNo} \rightarrow \text{CourseNo}$	Determinant not a superkey, but CourseNo is prime — exception applies ✓

Is Enrollment in BCNF?

Rule checked	Result
$\{\text{StudentNo}, \text{CourseNo}\} \rightarrow \text{InstructorNo}$	Determinant IS a candidate key ✓
$\text{InstructorNo} \rightarrow \text{CourseNo}$	Determinant NOT a superkey — VIOLATION, no exception exists

The relation is in 3NF, yet NOT in BCNF — and still carries real redundancy: I1 teaching CS201 repeats once per enrolled student.

- Modification — reassigning I1 to CS410 must update BOTH S1's and S2's rows together
- Insertion — instructor I3 (assigned CS500) can't be recorded until a student enrolls with them
- Deletion — S3 is I2's only enrollment; deleting it erases that I2 teaches CS305 at all

Testing Each FD Against BCNF

TEST

For each FD, is the determinant a superkey? Candidate keys: {StudentNo, CourseNo} and {StudentNo, InstructorNo}.

FD	Determinant X	X ⁺	X superkey?	Verdict
{StudentNo, CourseNo} → InstructorNo	{StudentNo, CourseNo}	all 3 attributes	yes	OK
InstructorNo → CourseNo	{InstructorNo}	{InstructorNo, CourseNo}	no	VIOLATES

WHY 3NF MISSED IT

CourseNo is a prime attribute (it lies in a candidate key), so 3NF’s exception applies. BCNF has no such exception.

BCNF Decomposition

$R_1 = X \cup Y$ (the violating FD becomes R_1 's own key)
 $R_2 = (R - Y) \cup X$ (everything else, keeping X to reconnect)

Applying it to Enrollment using the violating FD $\text{InstructorNo} \rightarrow \text{CourseNo}$ ($X=\{\text{InstructorNo}\}$, $Y=\{\text{CourseNo}\}$):

Teaches (InstructorNo, CourseNo)

InstructorNo	CourseNo
I1	CS201
I2	CS305

Enrolls (StudentNo, InstructorNo)

StudentNo	InstructorNo
S1	I1
S2	I1
S3	I2

TIP

Both are now in BCNF: Teaches's only determinant (InstructorNo) IS its own key. Enrolls has no non-trivial FD left to violate anything. I1's course now lives in exactly one row of Teaches — all three anomalies are gone.

The Decomposition Procedure, Step by Step

WHY

Each split follows one violating FD. Lecture 21 gives the full algorithm for relations with several violations.

Step	Action	Result for Enrollment
1	Find a non-trivial FD $X \rightarrow Y$ that violates BCNF	$\text{InstructorNo} \rightarrow \text{CourseNo}$
2	Form $R1 = X \cup Y$	$R1 = \text{Teaches}(\text{InstructorNo}, \text{CourseNo})$
3	Form $R2 = (R - Y) \cup X$	$R2 = \text{Enrolls}(\text{StudentNo}, \text{InstructorNo})$
4	Check the join: shared attributes must be a key of $R1$ or $R2$	$\{\text{InstructorNo}\}$ is the key of Teaches
5	Repeat on $R1$ and $R2$ until every relation is in BCNF	Both are in BCNF: stop

The Lossless-Join Test by Rule

RULE

R1 and R2 join losslessly if their shared attributes contain a key of R1 or of R2.

Check	Value	Result
Shared attributes $R1 \cap R2$	{InstructorNo}	—
Is {InstructorNo} a key of Teaches(InstructorNo, CourseNo)?	yes (its primary key)	Lossless ✓
Is {InstructorNo} a key of Enrolls(StudentNo, InstructorNo)?	no	Not needed

NOTE

Only one side needs the key. Teaches supplies it here, so the BCNF split is lossless.

Rejoining Teaches and Enrolls

JOIN

Teaches ⋈ Enrolls on InstructorNo, one Enrolls row at a time:

Enrolls row	Teaches row (same InstructorNo)	Joined row	In original?
(S1, I1)	(I1, CS201)	(S1, CS201, I1)	yes, row 1
(S2, I1)	(I1, CS201)	(S2, CS201, I1)	yes, row 2
(S3, I2)	(I2, CS305)	(S3, CS305, I2)	yes, row 3

RESULT

The join returns exactly the three original Enrollment rows: no rows lost, no spurious rows gained.

Lossless — But Not Dependency-Preserving

NOTE

A decomposition into R_1 and R_2 is lossless whenever the shared attributes $(R_1 \cap R_2)$ form a key of at least one relation: $(R_1 \cap R_2) \rightarrow R_1$ or $(R_1 \cap R_2) \rightarrow R_2$ holds in F^+ .

Teaches $\cap$ Enrolls = {InstructorNo}. InstructorNo is Teaches's declared PK, so $\text{InstructorNo} \rightarrow \text{Teaches}$ holds trivially.

Lossless: Teaches $\bowtie$ Enrolls (on InstructorNo) reconstructs Enrollment exactly.

WARNING

Check the ORIGINAL FD $\{\text{StudentNo}, \text{CourseNo}\} \rightarrow \text{InstructorNo}$ against the decomposed schema: StudentNo lives only in Enrolls, CourseNo only in Teaches — no single relation has both attributes, so this dependency cannot be verified without joining the tables back together. This is not a mistake — BCNF decomposition always guarantees lossless-join, but does NOT always guarantee dependency-preservation. 3NF (via synthesis, Lecture 21) guarantees both simultaneously — a genuine trade-off.

KEY TAKEAWAYS

Key Takeaways

- 1 Armstrong's Axioms — reflexivity, augmentation, transitivity — are sound and complete; union, decomposition, pseudotransitivity are convenient derived rules
- 2 Attribute closure (X^+) finds every attribute X determines, and is the mechanical way to find candidate keys
- 3 BCNF requires every non-trivial FD's determinant to be a superkey — no exception. 3NF tolerates one: a non-superkey determinant is OK if every attribute it determines is prime
- 4 Enrollment(StudentNo, CourseNo, InstructorNo) with $\text{InstructorNo} \rightarrow \text{CourseNo}$ is 3NF but NOT BCNF — and still carries real redundancy
- 5 BCNF decomposition splits along a violating FD ($R_1=XY$, $R_2=(R-Y)UX$) and is always lossless-join, but can sacrifice dependency-preservation